

Switching

In computer networking, **Switching** is the process of transferring data packets from one device to another in a network, or from one network to another, using specific devices called **switches**. A computer user experiences switching all the time for example, accessing the Internet from your computer device, whenever a user requests a webpage to open, the request is processed through switching of data packets only.

A. Circuit switching

Circuit Switching is a connection-oriented service. It provides a dedicated path from the sender to the receiver. In-circuit switching, a connection setup is required to send and receive data. It has very little chance of data loss and error due to the dedicated circuit, but a lot of bandwidth is wasted because the same path cannot be used by other senders during a congestion. Circuit switching is completely transparent; the sender and receiver can use any bit rate format or framing method.

Advantages of Circuit Switching

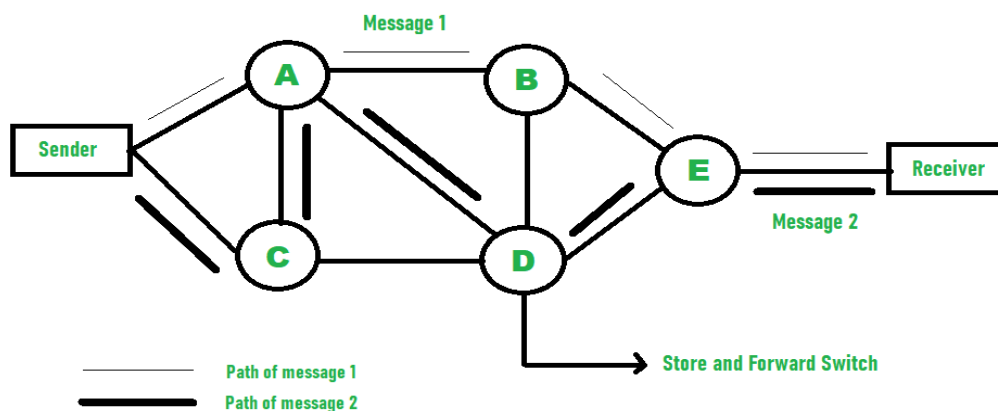
- It uses a fixed bandwidth.
- A dedicated communication channel increases the quality of communication.
- Data is transmitted with a fixed data rate.
- No waiting time at switches.
- Suitable for long continuous communication.

Disadvantages of circuit switching

- A dedicated connection makes it impossible to transmit other data even if the channel is free.
- Resources are not utilized fully.
- The time required to establish the physical link between the two stations is too long.
- A dedicated path has to be established for each connection.
- Circuit switching is more expensive.
- Even if there is no transfer of data, the link is still maintained until it is terminated by users.
- Dedicated channels require more bandwidth.

B. MESSAGE SWITCHING

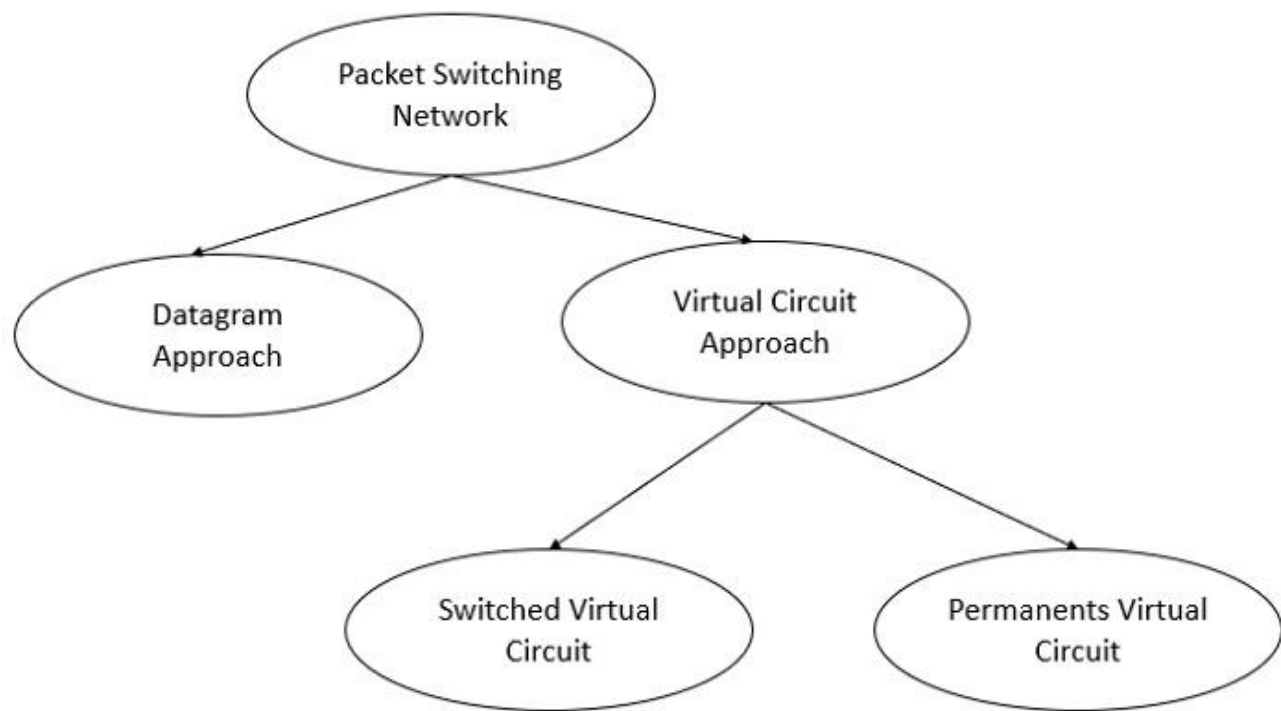
Switching is the technique by which nodes control or switch data to transmit it between specific points on a network. In message switching the entire message is transmitted without any break from one node to another. There is no direct link present between the sender and the receiver in message switching. Message switching is a switching mechanism in which a message is sent as a single unit and routed to intermediary nodes where it is stored and forwarded. The message-switching approach does not provide a dedicated path between the sender and receiver. In message switching, end-users communicate by sending and receiving *messages* that include the entire data to be shared. Messages are the smallest individual unit. Also, the sender and receiver are not directly connected. Several intermediate nodes transfer data and ensure that the message reaches its destination. Message-switched data networks are hence called hop-by-hop systems.



- **Store and forward** – The intermediate nodes have the responsibility of transferring the entire message to the next node. Hence, each node must have storage capacity. A message will only be delivered if the next hop and the link connecting it are both available, otherwise, it'll be stored indefinitely. A store-and-forward switch forwards a message only if sufficient resources are available and the next hop is accepting data. This is called the store-and-forward property.
- **Message delivery** – This implies wrapping the entire information in a single message and transferring it from the source to the destination node. Each message must have a header that contains the message routing information, including the source and destination.

Message switching network consists of transmission links (channels), store-and-forward switch nodes, and end stations

C. PACKET SWITCHING



Packet Switched service transfers the data from source to destination. Data is transferred on a type of network in which small units of data called packets are transferred. Each packet contains a destination address within it, where it has to be received.

This type of communication between receiver & sender is connectionless. The internet is also a connectionless network. Most of the traffic over the internet uses packet switched service.

Voice call over the internet is a type of packet switching, where each end of the conversation is divided into small packets & is reassembled later into a complete message. Packets are made up of a header and a payload, header directs it to the destination & data in the payload is the actual data which is to be transferred.

Types of Packet Switching Network

- Datagram packet switching
- Virtual circuit packet switching

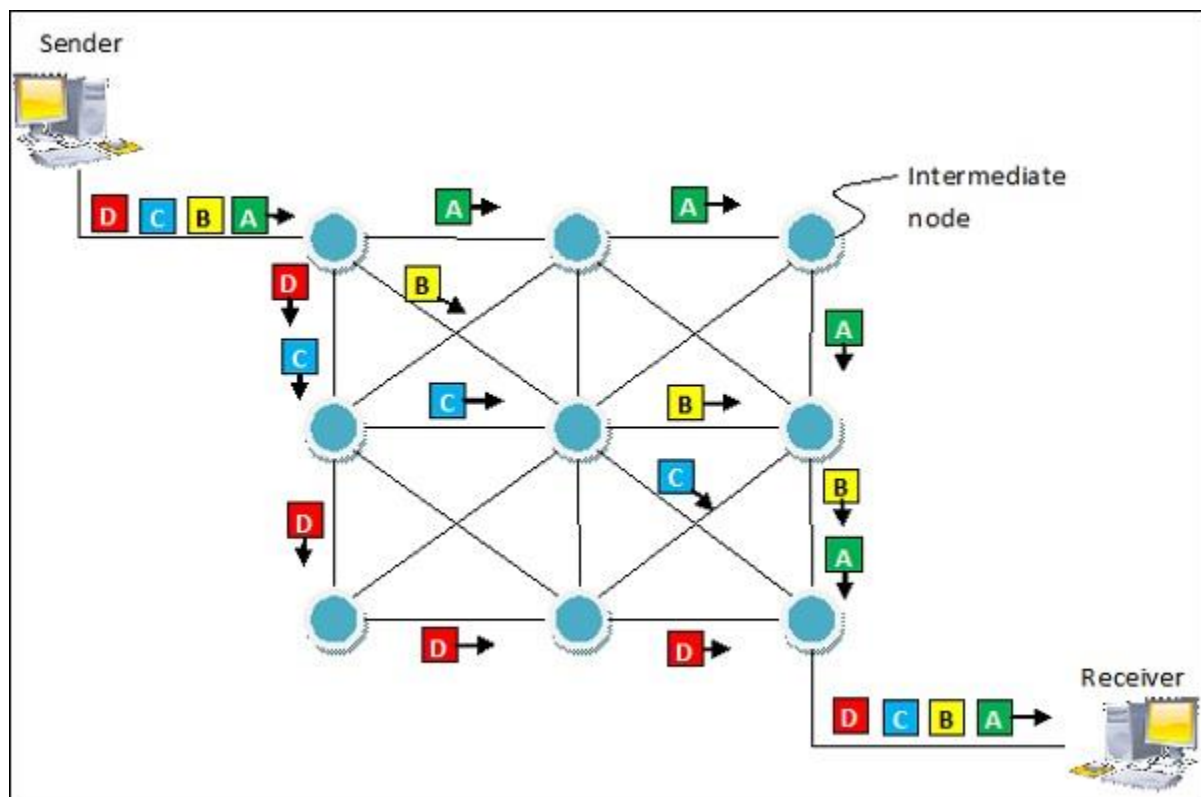
a. Datagram packet switching

Process

Each packet in a packet switching technique has two parts: a header and a payload. The header contains the addressing information of the packet and is used by the intermediate routers to direct it towards its destination. The payload carries the actual data.

A packet is transmitted as soon as it is available in a node, based upon its header information. The packets of a message are not routed via the same path. So, the packets in the message arrives in the destination out of order. It is the responsibility of the destination to reorder the packets in order to retrieve the original message.

The process is diagrammatically represented in the following figure. Here the message comprises of four packets, A, B, C and D, which may follow different routes from the sender to the receiver.



Advantages and Disadvantages of Packet Switching

Advantages

- Delay in delivery of packets is less, since packets are sent as soon as they are available.
- Switching devices don't require massive storage, since they don't have to store the entire messages before forwarding them to the next node.
- Data delivery can continue even if some parts of the network faces link failure. Packets can be routed via other paths.
- It allows simultaneous usage of the same channel by multiple users.
- It ensures better bandwidth usage as a number of packets from multiple sources can be transferred via the same link.

Disadvantages

- They are unsuitable for applications that cannot afford delays in communication like high quality voice calls.
- Packet switching high installation costs.
- They require complex protocols for delivery.
- Network problems may introduce errors in packets, delay in delivery of packets or loss of packets. If not properly handled, this may lead to loss of critical information.

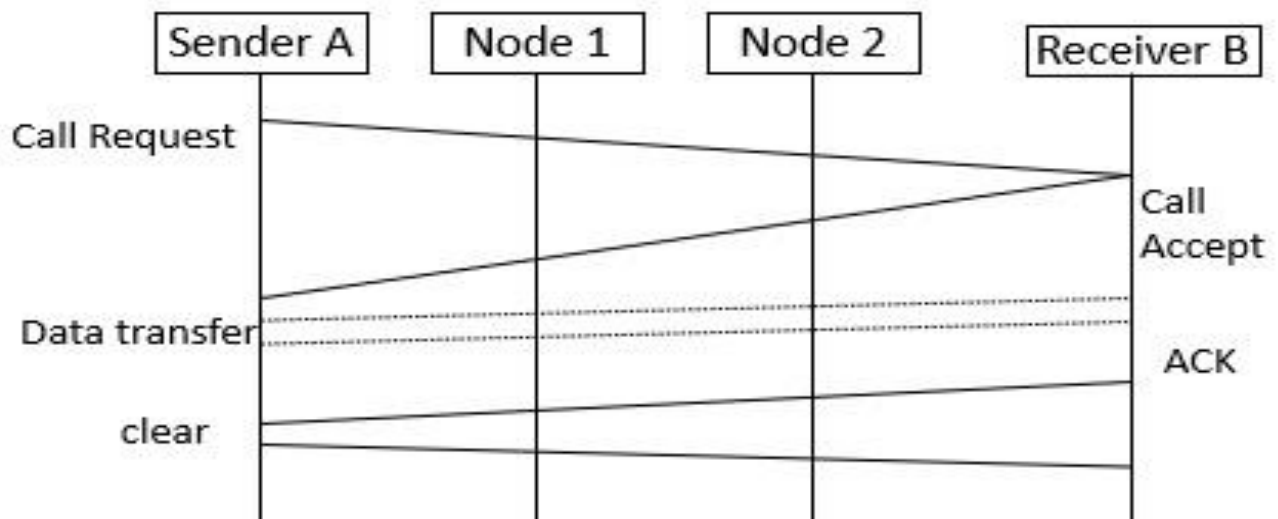
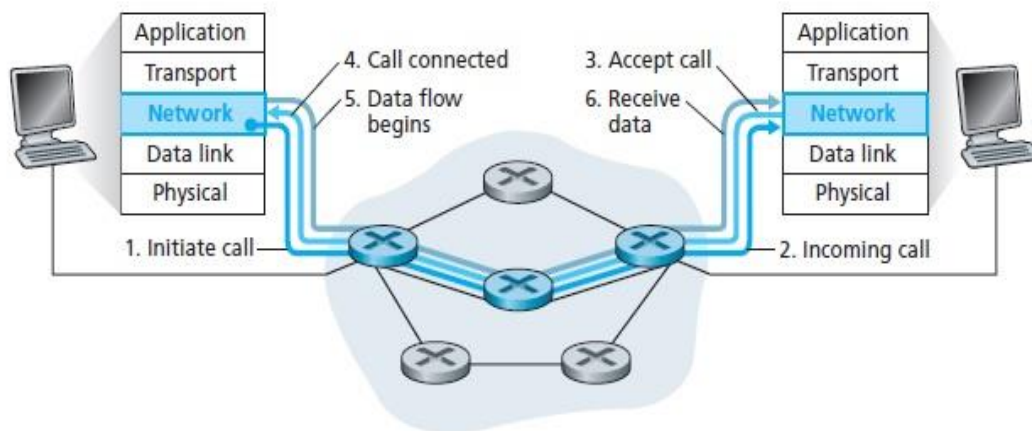
Difference between circuit switching and packet switching

Circuit Switching	Packet Switching
Circuit switching requires a dedicated path before sending data from source to destination.	Packet switching does not require any dedicated path to send data from source to destination.
It reserves the entire bandwidth in advance.	It does not reserve bandwidth in advance
No store and forward transmission	It does not reserve bandwidth in advance
Each packet follows the same route	A packet can follow any route
Call setup is required	No call setup is required
Bandwidth wastage	No bandwidth wastage

b.Virtual circuit packet switching

It is a network where a virtual connection is established between source and the destination. Through this network, packets will be transferred during any call. The path established between two points appears as a dedicated physical circuit. Therefore, it is called a virtual circuit. It is a type of packet switching.

It is a connection-oriented service, where the first packet goes and reserves the resources for the subsequent packets.



Explanation

Step 1 – Sender A establishes a call request connection to connect with the receiver.

Step 2 – Receiver B establishes a call accepting connection to connect with sender.

Step 3 – Data will be transferred whenever the router is established.

Step 4 – Node 1 and Node 2 are intermediate nodes between sender and receiver, the data will be transferred by connecting two nodes virtually.

Step 5 – After transmitting the data, an ACK will be sent by the receiver by saying a message is received.

Step 6 – A clear signal will be sent if the user wants to terminate the connection.

Advantages

The advantages of virtual circuit are as follows –

- Packets are delivered in the same order as they all follow the same route between the source & the destination.
- The overhead is smaller as full address is not required on each packet as they all follow the same established path.
- The connection is more reliable as it is one to one connection.
- Less chances of data loss.

Disadvantages

The disadvantages of virtual circuit are as follows –

- The switching equipment should be powerful.
- Re-establishment of the network is difficult if there is any failure. All calls need to be re-established.